Chapter 1 — Differentiation: Detailed Teaching Notes

Maharashtra HSC · Mathematics & Statistics (Arts & Science) Part II · Textbook pages 1–64 · File 1 of 3

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1 How to use these notes

Scope. This file covers the whole of Chapter 1 of the official Std XII Part II textbook (pages 1–64): Exercises 1.1 to 1.5 and Miscellaneous Exercise 1. Section numbers below match the textbook (1.1.1, 1.2.3, …) so you can jump between the book and these notes.

Textbook section	Topic	Exercise
1.1	Derivatives of composite functions (chain rule)	Ex. 1.1
1.2.1	Geometrical meaning of derivative	Ex. 1.1
1.2.2 – 1.2.6	Derivatives of inverse functions and inverse trigonometric functions	Ex. 1.2
1.3	Logarithmic differentiation; implicit functions	Ex. 1.3
1.4	Parametric functions; one function w.r.t. another	Ex. 1.4
1.5	Higher order derivatives; n^{th} derivatives	Ex. 1.5

Notation used by the textbook (and in these notes).

- $\log x$ means the natural logarithm $\log_e x$ (your textbook writes log, not ln).
- $y_1 = \dfrac{dy}{dx}$, $y_2 = \dfrac{d^2y}{dx^2}$, $y_n = \dfrac{d^ny}{dx^n}$. Also $f'(x), f''(x)$.
- "w.r.t." = with respect to. "Diff." = differentiating.

1.1 What the board actually asks from this chapter (verified)

I went through all six board papers in your project folder (March 2024, July 2024, February 2025, July 2025, February 2026, June 2026). Every question from Chapter 1 that appeared is listed here. Nothing below is guessed.

Paper (code)	Q. no.	Marks	Question (short form)	Topic
Mar 2024 (J-862)	Q1(v) MCQ	2	Slope of tangent to $x = \sin\theta,\ y = \cos 2\theta$ at $\theta = \frac{\pi}{6}$	Parametric
Mar 2024	Q9	2	$\frac{dy}{dx}$ if $y = (\log x)^x$	Log. diff.
Mar 2024	Q21	3	$y = \sin^{-1} x \Rightarrow (1-x^2)y_2 - xy_1 = 0$	2nd order
Mar 2024	Q31	4	Prove parametric rule; hence $x = at^2,\ y = 2at$	Proof + param.
Jul 2024 (J-174)	Q1(v) MCQ	2	$x = e^{x/y}$, find $\frac{dy}{dx}$	Implicit
Jul 2024	Q21	3	$y = \tan^{-1}\sqrt{\frac{3-x}{3+x}}$	Inv. trig subst.
Jul 2024	Q31	4	Prove chain rule; hence $\frac{d}{dx}\left(\frac{1}{\sqrt{\sin x}}\right)$	Proof + chain
Feb 2025 (J-312)	Q1(v) MCQ	2	$\frac{d}{dx}\sqrt{f^2+g^2}$ at $x=1$ from given values	Chain (values)
Feb 2025	Q21	3	n^{th} derivative of $\log x$	n^{th} deriv.
Feb 2025	Q31	4	Prove parametric rule; hence derivative of 7^x w.r.t. x^7	Proof + w.r.t.
Jul 2025 (J-384)	Q1(v) MCQ	2	$\log(x+y) = xy$, find $\frac{dy}{dx}$ at $x=0$	Implicit
Jul 2025	Q2(iii)	1	$\frac{dy}{dx}$ if $y = \sin(\log x)$	Chain
Jul 2025	Q9	2	$x\sqrt{x} + y\sqrt{y} = a\sqrt{a}$	Implicit
Jul 2025	Q31	4	Prove parametric rule; hence $y = at^2,\ x = 2at$	Proof + param.
Feb 2026 (J-165)	Q1(v) MCQ	2	$y = \sec(\tan^{-1} x)$, $\frac{dy}{dx}$ at $x=1$	Inv. trig
Feb 2026	Q9	2	$\sqrt{x} + \sqrt{y} = \sqrt{a}$	Implicit
Feb 2026	Q10	2	$\frac{d^2y}{dx^2}$ if $y = x^3 + 7x^2 - 2x - 9$	2nd order
Feb 2026	Q21	3	Prove chain rule	Proof
Jun 2026 (J-276)	Q1(v) MCQ	2	$y = \log_x a$, find $\frac{dy}{dx}$	Chain/log
Jun 2026	Q21	3	$y = \log\left[\frac{x+\sqrt{x^2+a^2}}{\sqrt{x^2+a^2}-x}\right]$	Log + simplify
Jun 2026	Q31	4	Prove parametric rule; hence $x = \sin t,\ y = \cos t$	Proof + param.

Exam Tip: What this pattern tells you

- **Every** paper had a 2-mark MCQ from this chapter at Q1(v).
- **Five of six** papers had a 4-mark Section D question that is a **proof (parametric rule or chain rule) + a "hence" sum**. The parametric proof appeared in 4 of 6 papers. Learn both proofs word-for-word (Sections 2 and 6).
- About **9–11 marks per paper** came from this chapter (before internal choice).
- Several MCQs were lifted almost directly from **Miscellaneous Exercise 1 Part (I)** (e.g. $\sec(\tan^{-1} x)$, $\sqrt{f^2+g^2}$). Do that exercise fully.
- This is a pattern from six papers, not a guarantee. Any textbook question can be asked.

2 Prerequisites you must already have (Let us Recall)

2.1 Derivative as a limit

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}, \qquad \frac{dy}{dx} = \lim_{\delta x\to 0} \frac{\delta y}{\delta x}$$

Here δx is a small change in x and δy the matching change in y. The derivative is the **rate at which y changes per unit change in x**.

2.2 Standard derivatives (Table 1.1.1 of textbook)

$y = f(x)$	$\frac{dy}{dx}$	$y = f(x)$	$\frac{dy}{dx}$
c (constant)	0	$\sec x$	$\sec x \tan x$
x^n	nx^{n-1}	$\operatorname{cosec} x$	$-\operatorname{cosec} x \cot x$
$\frac{1}{x}$	$-\frac{1}{x^2}$	$\cot x$	$-\operatorname{cosec}^2 x$
$\sqrt{x}$	$\frac{1}{2\sqrt{x}}$	e^x	e^x
$\sin x$	$\cos x$	a^x	$a^x \log a$
$\cos x$	$-\sin x$	$\log x$	$\frac{1}{x}$
$\tan x$	$\sec^2 x$	$\log_a x$	$\frac{1}{x \log a}$

Memory Trick: Signs of trig derivatives

Every trig function that starts with **"co"** (cos, cot, cosec) has a **minus** sign in its derivative. The other three (sin, tan, sec) are positive.

2.3 Rules of differentiation

If u and v are differentiable functions of x:

$$\frac{d}{dx}(u \pm v) = \frac{du}{dx} \pm \frac{dv}{dx}, \qquad \frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}, \qquad \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2} \ (v \neq 0)$$

Memory Trick: Quotient rule order

"**Low d-High minus High d-Low, over Low squared.**" The minus sign makes the order matter; product rule order does not.

3 1.1 Derivatives of Composite Functions (Chain Rule)

3.1 Intuition first

A composite function is a "function of a function": $\sin(x^2)$ means *first* square x, *then* take sine. Think of a set of gears. If gear u turns 3 times for every turn of gear x, and gear y turns 2 times for every turn of gear u, then y turns $2 \times 3 = 6$ times for each turn of x.

$$\underbrace{\frac{dy}{dx}}_{y \text{ per } x} = \underbrace{\frac{dy}{du}}_{y \text{ per } u} \times \underbrace{\frac{du}{dx}}_{u \text{ per } x}$$

Rates multiply along the chain. That is the whole idea.

Theorem: 1.1.2 Chain rule (textbook statement)

If $y = f(u)$ is a differentiable function of u and $u = g(x)$ is a differentiable function of x, then y is a differentiable function of x and

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}.$$

3.2 Proof (asked as 3 or 4 marks: Jul 2024 Q31, Feb 2026 Q21)

Proof: Write it exactly in this order

Given: $y = f(u)$ and $u = g(x)$. We assume u is not a constant function.
Let δx be a small increment in x, and let δu and δy be the corresponding increments in u and y, where $\delta x \neq 0, \delta u \neq 0, \delta y \neq 0$.
We have

$$\frac{\delta y}{\delta x} = \frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x}.$$

Taking the limit as $\delta x \to 0$ on both sides,

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta x \to 0} \left(\frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x} \right) = \lim_{\delta x \to 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \to 0} \frac{\delta u}{\delta x}.$$

As $\delta x \to 0$, we get $\delta u \to 0$ ($\because u$ is a continuous function of x, being differentiable).

$$\therefore \lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \lim_{\delta u \to 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \to 0} \frac{\delta u}{\delta x} \qquad \text{.....(I)}$$

Since y is a differentiable function of u and u is a differentiable function of x,

$$\lim_{\delta u \to 0} \frac{\delta y}{\delta u} = \frac{dy}{du} \quad \text{and} \quad \lim_{\delta x \to 0} \frac{\delta u}{\delta x} = \frac{du}{dx} \qquad \text{.....(II)}$$

From (I) and (II),

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy}{du} \cdot \frac{du}{dx} \qquad \text{.....(III)}$$

The R.H.S. of (III) exists and is finite, so the L.H.S. also exists and is finite. Hence $\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy}{dx}$ exists, i.e. y is a differentiable function of x, and

$$\boxed{\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}}$$

Exam Tip: Where marks go in this proof

1. Writing $\frac{\delta y}{\delta x} = \frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x}$ with the non-zero conditions.
2. The reason **"$\delta u \to 0$ as $\delta x \to 0$ because u is continuous"**. Students skip this; it is the key logical step.
3. Identifying both limits as derivatives, and concluding that the L.H.S. exists.

Extensions (Notes after the theorem).

- $\frac{d}{dx} f[g(x)] = f'[g(x)] \cdot g'(x)$.
- Three links: if $y = f(v)$, $v = g(u)$, $u = h(x)$, then $\frac{dy}{dx} = \frac{dy}{dv} \cdot \frac{dv}{du} \cdot \frac{du}{dx}$. Any number of links works the same way.

3.3 Standard composite derivatives (Table 1.1.2)

Read f as "the inside". The pattern is always: **derivative of outside (inside kept as it is) $\times$ derivative of inside.**

y	$\frac{dy}{dx}$	y	$\frac{dy}{dx}$
$[f(x)]^n$	$n[f(x)]^{n-1} f'(x)$	$\cot f(x)$	$-\operatorname{cosec}^2 f(x) \cdot f'(x)$
$\sqrt{f(x)}$	$\frac{f'(x)}{2\sqrt{f(x)}}$	$\sec f(x)$	$\sec f(x) \tan f(x) \cdot f'(x)$
$\frac{1}{[f(x)]^n}$	$-\frac{n f'(x)}{[f(x)]^{n+1}}$	$\operatorname{cosec} f(x)$	$-\operatorname{cosec} f(x) \cot f(x) \cdot f'(x)$
$\sin f(x)$	$\cos f(x) \cdot f'(x)$	$e^{f(x)}$	$e^{f(x)} \cdot f'(x)$
$\cos f(x)$	$-\sin f(x) \cdot f'(x)$	$a^{f(x)}$	$a^{f(x)} \log a \cdot f'(x)$
$\tan f(x)$	$\sec^2 f(x) \cdot f'(x)$	$\log f(x)$	$\frac{f'(x)}{f(x)}$
		$\log_a f(x)$	$\frac{f'(x)}{f(x) \log a}$

3.4 Worked examples

Worked Example: 1 (Easy) — $y = \sqrt{x^2+5}$

Method 1 (substitution, textbook shows this first). Let $u = x^2+5$, so $y = \sqrt{u}$.

$$\frac{dy}{du} = \frac{1}{2\sqrt{u}}, \qquad \frac{du}{dx} = 2x$$

$$\therefore \frac{dy}{dx} = \frac{dy}{du}\cdot\frac{du}{dx} = \frac{1}{2\sqrt{x^2+5}}\cdot 2x = \frac{x}{\sqrt{x^2+5}}$$

Method 2 (direct, what you should use from now on).

$$\frac{dy}{dx} = \frac{1}{2\sqrt{x^2+5}}\cdot\frac{d}{dx}(x^2+5) = \frac{2x}{2\sqrt{x^2+5}} = \frac{x}{\sqrt{x^2+5}}$$

Worked Example: 2 (Easy) — $y = \sin(\log x)$ [Board, Jul 2025 Q2(iii), 1 mark]

$$\frac{dy}{dx} = \cos(\log x)\cdot\frac{d}{dx}(\log x) = \cos(\log x)\cdot\frac{1}{x} = \frac{\cos(\log x)}{x}$$

Worked Example: 3 (Medium) — $y = 5^{3\cos x - 2}$

Outer function is a^u with $a = 5$.

$$\frac{dy}{dx} = 5^{3\cos x-2}\cdot\log 5\cdot\frac{d}{dx}(3\cos x - 2) = 5^{3\cos x-2}\cdot\log 5\cdot(-3\sin x) = -3\sin x\cdot 5^{3\cos x-2}\log 5$$

Worked Example: 4 (Medium) — three layers: $y = \log[\cos(x^3-5)]$

Layers from outside: $\log(\,\cdot\,)$, then $\cos(\,\cdot\,)$, then x^3-5.

$$\frac{dy}{dx} = \frac{1}{\cos(x^3-5)}\cdot\frac{d}{dx}\cos(x^3-5) = \frac{1}{\cos(x^3-5)}\cdot[-\sin(x^3-5)]\cdot 3x^2 = -3x^2\tan(x^3-5)$$

Worked Example: 5 (Medium) — $y = \log_x a$ [Board, Jun 2026 Q1(v) MCQ]

Key step: change the base first. $\log_x a = \dfrac{\log a}{\log x}$, and $\log a$ is a constant.

$$y = \log a\cdot(\log x)^{-1}$$

$$\frac{dy}{dx} = \log a\cdot(-1)(\log x)^{-2}\cdot\frac{1}{x} = \frac{-\log a}{x(\log x)^2}$$

Correct option: (a).

Worked Example: 6 (Hard) — using a value table: textbook Solved Ex. 4 and 5

(a) If $f(x) = \sqrt{7g(x)-3}$, $g(3) = 4$ and $g'(3) = 5$, find $f'(3)$.

$$f'(x) = \frac{1}{2\sqrt{7g(x)-3}}\cdot 7g'(x) \quad\Rightarrow\quad f'(3) = \frac{7g'(3)}{2\sqrt{7g(3)-3}} = \frac{7(5)}{2\sqrt{28-3}} = \frac{35}{10} = \frac{7}{2}$$

(b) If $F(x) = G\{3G[5G(x)]\}$, $G(0) = 0$ and $G'(0) = 3$, find $F'(0)$.

$$F'(x) = G'\{3G[5G(x)]\}\cdot 3G'[5G(x)]\cdot 5G'(x)$$

At $x=0$: $5G(0)=0$, so $G'[5G(0)]=G'(0)=3$; and $3G[5G(0)]=3G(0)=0$, so $G'\{\dots\}=G'(0)=3$.

$$F'(0)=3\cdot(3\cdot3)\cdot(5\cdot3)=3\cdot9\cdot15=405$$

Worked Example: 7 (Hard) — Board Feb 2025 Q1(v) / Misc. Ex. 1 (I)(1)

$f(1)=3$, $f'(1)=-\frac{1}{3}$, $g(1)=-4$, $g'(1)=-\frac{8}{3}$. Find $\frac{d}{dx}\sqrt{[f(x)]^2+[g(x)]^2}$ at $x=1$.

$$\frac{d}{dx}\sqrt{f^2+g^2}=\frac{1}{2\sqrt{f^2+g^2}}(2ff'+2gg')=\frac{ff'+gg'}{\sqrt{f^2+g^2}}$$

At $x=1$: $\sqrt{9+16}=5$, and $ff'+gg'=3\left(-\frac{1}{3}\right)+(-4)\left(-\frac{8}{3}\right)=-1+\frac{32}{3}=\frac{29}{3}$.

$$\text{Required value}=\frac{29/3}{5}=\frac{29}{15}\qquad\text{Option (D)}$$

Worked Example: 8 (Hard) — Ex. 1.1 Q7: where is $\frac{dy}{dx}=0$?

Find the x-coordinates of all points on $y=\sin 2x-2\sin x$, $0\le x<2\pi$, where $\frac{dy}{dx}=0$.

$$\frac{dy}{dx}=2\cos 2x-2\cos x=0\ \Rightarrow\ 2\cos^2 x-1-\cos x=0\ \Rightarrow\ (2\cos x+1)(\cos x-1)=0$$

$\cos x=1\Rightarrow x=0$; $\cos x=-\frac{1}{2}\Rightarrow x=\frac{2\pi}{3},\frac{4\pi}{3}$. Answer: $x=0,\ \frac{2\pi}{3},\ \frac{4\pi}{3}$.

Common Mistake: Chain rule errors that cost marks

- **Forgetting the inside derivative:** $\frac{d}{dx}\sin(x^2)=\cos(x^2)\times$. Correct: $2x\cos(x^2)$.
- **Changing the inside:** $\frac{d}{dx}\cos(3x)=-\sin x\cdot 3\times$. The inside stays $3x$.
- $\sin^2 x$ **vs** $\sin x^2$**:** $\sin^2 x=(\sin x)^2$, outermost layer is the square: derivative $2\sin x\cos x$. $\sin x^2=\sin(x^2)$: derivative $2x\cos x^2$.
- **Log of a constant base:** in $\log_x a$, the variable is in the base — always convert to $\frac{\log a}{\log x}$ first.

4 1.2.1 Geometrical Meaning of the Derivative

Take a point $P(a,f(a))$ on the curve and a nearby point $Q(a+h,f(a+h))$.

$$\text{Slope of secant } PQ=\frac{f(a+h)-f(a)}{h}$$

As $h\to 0$, Q slides along the curve towards P and the secant turns into the **tangent at** P. So

$$\text{Slope of tangent at } P=\lim_{h\to0}\frac{f(a+h)-f(a)}{h}=f'(a)\quad\text{(if the limit exists).}$$

Definition: Geometric meaning

$\left(\frac{dy}{dx}\right)_{(x_1,y_1)}$ is the **slope (gradient) of the tangent** to the curve $y=f(x)$ at the point (x_1,y_1).

Worked Example: Slope at a point

For $y=x^2+3x+2$, slope of tangent at $(2,12)$: $\frac{dy}{dx}=2x+3$, so $m=2(2)+3=7$. (Check the point is on the curve: $4+6+2=12$. ✓ Always check this in "at the point" questions.)

This idea is used heavily in Chapter 2 (tangents and normals). In this chapter it appears mainly in parametric MCQs like "slope of tangent at $\theta = \frac{\pi}{6}$".

5 1.2.2 – 1.2.3 Derivatives of Inverse Functions

5.1 Intuition

If $y = f(x)$ stretches x by a factor of 2 near a point (slope 2), the inverse function has to *undo* that, so it shrinks by a factor of 2 (slope $\frac{1}{2}$). Graphically, the graph of f^{-1} is the mirror image of f in the line $y = x$; mirroring swaps rise and run, so slopes become reciprocals.

Textbook example: $f(x) = 2x - 2$, $f^{-1}(x) = \frac{x+2}{2}$; derivatives 2 and $\frac{1}{2}$ — reciprocals.

Theorem: 1.2.3 Derivative of an inverse function

If $y = f(x)$ is a differentiable function of x such that $\frac{dy}{dx} \neq 0$ and $x = f^{-1}(y)$ exists, then $x = f^{-1}(y)$ is a differentiable function of y and

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}, \quad \text{where } \frac{dy}{dx} \neq 0.$$

Proof: Proof (textbook method)

Given $y = f(x)$ and $x = f^{-1}(y)$ are differentiable. Let δx be a small increment in x and δy the corresponding increment in y, with $\delta x \neq 0$, $\delta y \neq 0$.

$$\frac{\delta x}{\delta y} \times \frac{\delta y}{\delta x} = 1 \quad \therefore \quad \frac{\delta x}{\delta y} = \frac{1}{\frac{\delta y}{\delta x}}, \quad \frac{\delta y}{\delta x} \neq 0$$

Taking the limit as $\delta x \to 0$; as $\delta x \to 0$, $\delta y \to 0$ ($\because y$ is continuous):

$$\lim_{\delta y \to 0} \frac{\delta x}{\delta y} = \frac{1}{\lim\limits_{\delta x \to 0} \frac{\delta y}{\delta x}} \qquad \text{.....(I)}$$

Since y is differentiable, $\lim\limits_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy}{dx}$ and $\frac{dy}{dx} \neq 0$ .....(II)

From (I) and (II), $\lim\limits_{\delta y \to 0} \frac{\delta x}{\delta y} = \frac{1}{dy/dx}$ exists and is finite. Hence x is a differentiable function of y and

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}.$$

Alternative (chain rule). $f^{-1}[f(x)] = x$. Differentiating, $(f^{-1})'[f(x)] \cdot f'(x) = 1$, so $(f^{-1})'(y) = \frac{1}{f'(x)}$.

Useful form for MCQs: if g is the inverse of f, then $f[g(x)] = x$, so

$$g'(x) = \frac{1}{f'[g(x)]}.$$

5.2 Worked examples

Worked Example: 1 (Easy) — derivative of $y = \log x$ via its inverse

$y = \log x \Rightarrow x = e^y$. So $\frac{dx}{dy} = e^y$.

$$\frac{dy}{dx} = \frac{1}{dx/dy} = \frac{1}{e^y} = \frac{1}{e^{\log x}} = \frac{1}{x}$$

Worked Example: 2 (Medium) — derivative of the inverse at a point

Find the derivative of the inverse of $y = 2x^3 - 6x$ and its value at $x = -2$.

$$\frac{dy}{dx} = 6x^2 - 6 \quad \Rightarrow \quad \frac{dx}{dy} = \frac{1}{6(x^2-1)}$$

At $x = -2$: $\frac{dx}{dy} = \frac{1}{6(4-1)} = \frac{1}{18}$.

Worked Example: 3 (Hard) — $(f^{-1})'(-3)$ for $f(x) = x^5 + 2x - 3$ [textbook Solved Ex. 4]

Trap: you need the x for which $f(x) = -3$, *not* $x = -3$.
$f(x) = -3 \Rightarrow x^5 + 2x = 0 \Rightarrow x(x^4+2) = 0 \Rightarrow x = 0$.
$f'(x) = 5x^4 + 2 \Rightarrow f'(0) = 2$.

$$(f^{-1})'(-3) = \frac{1}{f'(0)} = \frac{1}{2}$$

Worked Example: 4 (Hard) — from a table [textbook Solved Ex. 3]

f, g are inverses. Given $g(-4) = 1$ and $f'(1) = 4$. Then

$$g'(-4) = \frac{1}{f'[g(-4)]} = \frac{1}{f'(1)} = \frac{1}{4}.$$

6 1.2.4 – 1.2.6 Derivatives of Inverse Trigonometric Functions

6.1 Derivation of $\frac{d}{dx}(\sin^{-1} x)$ — learn this method

Let $y = \sin^{-1} x$, $-1 < x < 1$, $-\frac{\pi}{2} < y < \frac{\pi}{2}$. Then $x = \sin y$.

$$\frac{dx}{dy} = \cos y$$

In $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\cos y > 0$, so $\cos y = +\sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}$.

$$\therefore \frac{dy}{dx} = \frac{1}{dx/dy} = \frac{1}{\sqrt{1-x^2}}, \qquad |x| < 1$$

The others are derived identically (try $\tan^{-1} x$: $x = \tan y$, $\frac{dx}{dy} = \sec^2 y = 1 + x^2$).

Formula: Inverse trig derivatives (textbook table)

$f(x)$	$f'(x)$	Condition
$\sin^{-1} x$	$\dfrac{1}{\sqrt{1-x^2}}$	$\lvert x\rvert < 1$
$\cos^{-1} x$	$-\dfrac{1}{\sqrt{1-x^2}}$	$\lvert x\rvert < 1$
$\tan^{-1} x$	$\dfrac{1}{1+x^2}$	$x \in \mathbb{R}$
$\cot^{-1} x$	$-\dfrac{1}{1+x^2}$	$x \in \mathbb{R}$
$\sec^{-1} x$	$\dfrac{1}{x\sqrt{x^2-1}}$	$\lvert x\rvert > 1$
$\operatorname{cosec}^{-1} x$	$-\dfrac{1}{x\sqrt{x^2-1}}$	$\lvert x\rvert > 1$

Composite forms: replace x by $f(x)$ and multiply by $f'(x)$. E.g. $\dfrac{d}{dx}\tan^{-1} f(x) = \dfrac{f'(x)}{1+[f(x)]^2}$.

Memory Trick: Remembering the six

- Pairs are "co-partners": $\sin^{-1}$ & $\cos^{-1}$, $\tan^{-1}$ & $\cot^{-1}$, $\sec^{-1}$ & $\operatorname{cosec}^{-1}$. Each "co" version is the **negative** of its partner.
- Why: $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ (a constant), so their derivatives must add to zero. Same for the other two pairs.
- $\sqrt{1-x^2}$ goes with sine/cosine (circle $x^2+y^2=1$); $1+x^2$ goes with tangent ($1+\tan^2 = \sec^2$); $x\sqrt{x^2-1}$ goes with secant.

Note: Textbook remark on $\sec^{-1} x$

The textbook uses $\dfrac{1}{x\sqrt{x^2-1}}$. (Higher books write $\dfrac{1}{\lvert x\rvert\sqrt{x^2-1}}$; for board answers follow the textbook form.)

Worked Example: Using derivatives to prove an identity — $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ [Ex. 1.2 Q5 type]

Let $f(x) = \sin^{-1} x + \cos^{-1} x$. Then $f'(x) = \dfrac{1}{\sqrt{1-x^2}} - \dfrac{1}{\sqrt{1-x^2}} = 0$, so f is constant. Put $x = 0$: $f(0) = 0 + \frac{\pi}{2} = \frac{\pi}{2}$. Hence $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$.

6.2 Direct composite examples

Worked Example: 1 (Easy)

(a) $y = \tan^{-1}(\log x)$: $\dfrac{dy}{dx} = \dfrac{1}{1+(\log x)^2}\cdot\dfrac{1}{x} = \dfrac{1}{x[1+(\log x)^2]}$

(b) $y = \cot^{-1}(x^3)$: $\dfrac{dy}{dx} = -\dfrac{3x^2}{1+x^6}$

(c) $y = \cos^{-1}(1-x^2)$: $\dfrac{dy}{dx} = -\dfrac{1}{\sqrt{1-(1-x^2)^2}}\cdot(-2x) = \dfrac{2x}{\sqrt{2x^2-x^4}} = \dfrac{2}{\sqrt{2-x^2}}\ (x>0)$

Worked Example: 2 (Medium) — $y = \sec(\tan^{-1} x)$ [Board Feb 2026 Q1(v); Misc. Ex. (I)(2)]

Method A (chain rule).

$$\frac{dy}{dx} = \sec(\tan^{-1} x)\tan(\tan^{-1} x)\cdot\frac{1}{1+x^2} = \sec(\tan^{-1} x)\cdot\frac{x}{1+x^2}$$

At $x = 1$: $\tan^{-1} 1 = \frac{\pi}{4}$, $\sec\frac{\pi}{4} = \sqrt{2}$, so $\frac{dy}{dx} = \sqrt{2}\cdot\frac{1}{2} = \frac{1}{\sqrt{2}}$.

Method B (simplify first). Let $\tan^{-1} x = \theta$, so $\tan\theta = x$ and $\sec\theta = \sqrt{1+x^2}$ (positive, since $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$). So $y = \sqrt{1+x^2}$ and $\frac{dy}{dx} = \frac{x}{\sqrt{1+x^2}}$, which is $\frac{1}{\sqrt{2}}$ at $x = 1$. Option (C).

6.3 The substitution technique (most important part of Ex. 1.2)

When the inside of an inverse trig function is a messy algebraic expression, **don't differentiate it directly**. Substitute x = a trig function so the expression collapses into a single trig ratio; the inverse function then cancels.

Formula: Standard substitutions

Expression contains	Put	Collapses to
$\sqrt{1-x^2}$	$x = \sin\theta$ or $x = \cos\theta$	$\cos\theta$ or $\sin\theta$
$1+x^2$, $\frac{2x}{1-x^2}$, $\frac{2x}{1+x^2}$, $\frac{1-x^2}{1+x^2}$	$x = \tan\theta$	$\tan 2\theta$, $\sin 2\theta$, $\cos 2\theta$
$\sqrt{x^2-1}$	$x = \sec\theta$	$\tan\theta$
$3x - 4x^3$ / $4x^3 - 3x$	$x = \sin\theta$ / $x = \cos\theta$	$\sin 3\theta$ / $\cos 3\theta$
$\frac{3x-x^3}{1-3x^2}$	$x = \tan\theta$	$\tan 3\theta$
$\sqrt{\frac{a-x}{a+x}}$	$x = a\cos 2\theta$	$\tan\theta$
$\frac{1-\cos x}{1+\cos x}$, $\frac{\cos x}{1+\sin x}$ etc.	half-angle identities	$\tan^2\frac{x}{2}$, $\tan\left(\frac{\pi}{4}-\frac{x}{2}\right)$
$\frac{a\cos x + b\sin x}{\sqrt{a^2+b^2}}$	$\frac{a}{\sqrt{a^2+b^2}} = \cos\alpha$ (or $\sin\alpha$)	$\cos(x-\alpha)$ or $\sin(x+\alpha)$
$\frac{A\pm B}{1\mp AB}$ form	split: $\tan^{-1} A \pm \tan^{-1} B$	sum/difference

Warning: The domain condition — write it in the answer

$\sin^{-1}(\sin\phi) = \phi$ **only** if $\phi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$; $\cos^{-1}(\cos\phi) = \phi$ only if $\phi \in [0, \pi]$; $\tan^{-1}(\tan\phi) = \phi$ only if $\phi \in (-\frac{\pi}{2}, \frac{\pi}{2})$. Textbook solutions state a range for x (e.g. "for $0 < x < 1$") when cancelling. If a question gives a range, use it; if not, state a reasonable one. The answer can change sign in a different range — e.g. $\cos^{-1}\frac{1-x^2}{1+x^2}$ is $2\tan^{-1} x$ for $x \geq 0$ but $-2\tan^{-1} x$ for $x < 0$.

Worked Example: 3 (Medium) — $y = \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right),\ x > 0$

Put $x = \tan\theta$, so $\theta = \tan^{-1} x$ and $\theta \in \left(0, \frac{\pi}{2}\right)$.

$$\frac{1-x^2}{1+x^2} = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos 2\theta$$

$y = \cos^{-1}(\cos 2\theta) = 2\theta$ (valid since $2\theta \in (0, \pi)$) $= 2\tan^{-1} x$.

$$\frac{dy}{dx} = \frac{2}{1+x^2}$$

Worked Example: 4 (Medium) — $y = \sin^{-1}(2x\sqrt{1-x^2}),\ |x| < \frac{1}{\sqrt{2}}$

Put $x = \sin\theta$, $\theta = \sin^{-1} x \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$. Then $\sqrt{1-x^2} = \cos\theta$.

$$2x\sqrt{1-x^2} = 2\sin\theta\cos\theta = \sin 2\theta \ \Rightarrow\ y = \sin^{-1}(\sin 2\theta) = 2\theta = 2\sin^{-1} x$$

$$\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}$$

Worked Example: 5 (Medium) — half-angle: $y = \tan^{-1}\left(\dfrac{\cos x}{1+\sin x}\right)$

Write $\cos x = \cos^2\frac{x}{2} - \sin^2\frac{x}{2}$ and $1 + \sin x = \left(\cos\frac{x}{2} + \sin\frac{x}{2}\right)^2$.

$$\frac{\cos x}{1+\sin x} = \frac{(\cos\frac{x}{2} - \sin\frac{x}{2})(\cos\frac{x}{2} + \sin\frac{x}{2})}{(\cos\frac{x}{2} + \sin\frac{x}{2})^2} = \frac{\cos\frac{x}{2} - \sin\frac{x}{2}}{\cos\frac{x}{2} + \sin\frac{x}{2}} = \frac{1-\tan\frac{x}{2}}{1+\tan\frac{x}{2}} = \tan\left(\frac{\pi}{4} - \frac{x}{2}\right)$$

$$y = \frac{\pi}{4} - \frac{x}{2} \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{1}{2}$$

Worked Example: 6 (Hard) — $y = \tan^{-1}\sqrt{\dfrac{3-x}{3+x}}$ **[Board Jul 2024 Q21, 3 marks]**

Put $x = 3\cos 2\theta$, so $\theta = \frac{1}{2}\cos^{-1}\left(\frac{x}{3}\right)$. For $-3 < x < 3$, $2\theta \in (0, \pi)$ so $\theta \in \left(0, \frac{\pi}{2}\right)$.

$$\sqrt{\frac{3-x}{3+x}} = \sqrt{\frac{3-3\cos 2\theta}{3+3\cos 2\theta}} = \sqrt{\frac{2\sin^2\theta}{2\cos^2\theta}} = \tan\theta \quad (\tan\theta > 0)$$

$$y = \tan^{-1}(\tan\theta) = \theta = \frac{1}{2}\cos^{-1}\left(\frac{x}{3}\right)$$

Differentiating w.r.t. x:

$$\frac{dy}{dx} = \frac{1}{2}\cdot\frac{-1}{\sqrt{1-\frac{x^2}{9}}}\cdot\frac{1}{3} = \frac{-1}{6}\cdot\frac{3}{\sqrt{9-x^2}} = -\frac{1}{2\sqrt{9-x^2}}$$

Worked Example: 7 (Hard) — the $a\cos x + b\sin x$ form: $y = \sin^{-1}\left(\dfrac{4\sin x + 5\cos x}{\sqrt{41}}\right)$

Note $4^2 + 5^2 = 41$. Let $\dfrac{4}{\sqrt{41}} = \cos\alpha$, $\dfrac{5}{\sqrt{41}} = \sin\alpha$ (valid as $\cos^2\alpha + \sin^2\alpha = 1$).

$$\frac{4\sin x + 5\cos x}{\sqrt{41}} = \sin x\cos\alpha + \cos x\sin\alpha = \sin(x+\alpha)$$

$y = \sin^{-1}[\sin(x+\alpha)] = x + \alpha$ (for $x + \alpha$ in the principal range), and α is a constant.

$$\frac{dy}{dx} = 1$$

Worked Example: 8 (Hard) — splitting: $y = \tan^{-1}\left(\dfrac{8x}{1-15x^2}\right)$

Look for two numbers whose **sum is 8** and **product is 15**: 5 and 3.

$$\frac{8x}{1-15x^2} = \frac{5x+3x}{1-(5x)(3x)} \Rightarrow y = \tan^{-1}5x + \tan^{-1}3x$$

$$\frac{dy}{dx} = \frac{5}{1+25x^2} + \frac{3}{1+9x^2}$$

Similarly $\cot^{-1}\left(\dfrac{1+35x^2}{2x}\right) = \tan^{-1}\left(\dfrac{2x}{1+35x^2}\right) = \tan^{-1}\left(\dfrac{7x-5x}{1+7x\cdot 5x}\right) = \tan^{-1}7x - \tan^{-1}5x$, with derivative $\dfrac{7}{1+49x^2} - \dfrac{5}{1+25x^2}$.

Common Mistake: Inverse trig traps

- $\sin^{-1}x \neq \dfrac{1}{\sin x}$. The -1 means inverse function, not reciprocal.
- Differentiating $\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ directly with the quotient rule is legal but long and error-prone; substitution gives $2\tan^{-1}x$ in one line.
- After substituting, **re-express θ in terms of x before differentiating** (e.g. $\theta = \tan^{-1}x$). Differentiating "2θ" w.r.t. x as if it were 2 is a classic blunder.

7 1.3.1 Logarithmic Differentiation

7.1 When and why

Use it in two situations:

1. **Variable in both base and power**: x^x, $(\sin x)^{\tan x}$, $(\log x)^x$. No ordinary rule works here: x^n needs a constant power, a^x needs a constant base.
2. **Long products/quotients/powers**: e.g. $\dfrac{(x+1)^2(x+2)^3}{(x+3)^4}$. Logs turn products into sums, which are easy to differentiate.

Method. Take log of both sides → use log laws to bring powers down → differentiate both sides w.r.t. x (left side gives $\frac{1}{y}\frac{dy}{dx}$) → multiply by y → substitute back the original y.

Formula: Log laws you need

$$\log(mn) = \log m + \log n, \quad \log\frac{m}{n} = \log m - \log n, \quad \log m^n = n\log m, \quad \log_a b = \frac{\log b}{\log a}, \quad e^{\log x} = x$$

Formula: Textbook shortcut (Let us Remember)

$$\frac{d}{dx}(f^g) = f^g\left[\frac{g}{f}\cdot f' + (\log f)\cdot g'\right]$$

Read it as: "**treat the power as constant** ($gf^{g-1}f'$) **plus treat the base as constant** ($f^g \log f \cdot g'$)". Good for checking MCQs; in long answers show the log steps.

Worked Example: 1 (Easy) — $y = x^x$

$\log y = x\log x$. Differentiating w.r.t. x:

$$\frac{1}{y}\frac{dy}{dx} = x\cdot\frac{1}{x} + \log x\cdot 1 = 1 + \log x \quad \Rightarrow \quad \frac{dy}{dx} = x^x(1+\log x)$$

Worked Example: 2 (Medium) — $y = (\log x)^x$ [Board Mar 2024 Q9, 2 marks]

$\log y = x\log(\log x)$. Differentiating w.r.t. x:

$$\frac{1}{y}\frac{dy}{dx} = x\cdot\frac{1}{\log x}\cdot\frac{1}{x} + \log(\log x)\cdot 1 = \frac{1}{\log x} + \log(\log x)$$

$$\therefore \ \frac{dy}{dx} = (\log x)^x\left[\log(\log x) + \frac{1}{\log x}\right]$$

Worked Example: 3 (Medium) — $y = (\sin x)^{\tan x}$

$\log y = \tan x\cdot\log(\sin x)$.

$$\frac{1}{y}\frac{dy}{dx} = \tan x\cdot\frac{\cos x}{\sin x} + \log(\sin x)\cdot\sec^2 x = 1 + \sec^2 x\log(\sin x)$$

$$\frac{dy}{dx} = (\sin x)^{\tan x}\left[1 + \sec^2 x\log(\sin x)\right]$$

Worked Example: 4 (Hard) — a sum of two power-functions: $y = x^x + (\log x)^{\sin x}$

You cannot take log of a sum directly ($\log(u+v) \neq \log u + \log v$). Split: $y = u + v$, $\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$.

$u = x^x \Rightarrow \dfrac{du}{dx} = x^x(1+\log x)$ (Example 1).

$v = (\log x)^{\sin x} \Rightarrow \log v = \sin x\cdot\log(\log x)$:

$$\frac{1}{v}\frac{dv}{dx} = \sin x\cdot\frac{1}{x\log x} + \cos x\cdot\log(\log x)$$

$$\frac{dy}{dx} = x^x(1+\log x) + (\log x)^{\sin x}\left[\frac{\sin x}{x\log x} + \cos x\log(\log x)\right]$$

Worked Example: 5 (Hard) — product of many factors: $y = (x+1)^2(x+2)^3(x+3)^4$

$\log y = 2\log(x+1) + 3\log(x+2) + 4\log(x+3)$

$$\frac{dy}{dx} = (x+1)^2(x+2)^3(x+3)^4\left[\frac{2}{x+1} + \frac{3}{x+2} + \frac{4}{x+3}\right]$$

Common Mistake: Log differentiation errors

- Forgetting to multiply by y at the end, or leaving the answer as "$y[\ldots]$" without substituting y back. Write the full expression.
- Taking log of a sum (see Example 4).
- $\frac{d}{dx}\log(\log x) = \frac{1}{\log x}\cdot\frac{1}{x}$ — students drop the $\frac{1}{x}$.

8 1.3.2 – 1.3.3 Implicit Functions

8.1 Idea

An **explicit** function has y alone on one side: $y = x^2 + \sin x$. An **implicit** function mixes x and y: $x^3 + y^3 = 3axy$, where y is still a function of x but you cannot (easily) isolate it.

Method. Differentiate every term w.r.t. x, remembering y is a function of x, so every y-term gets an extra $\frac{dy}{dx}$ (chain rule):

$$\frac{d}{dx}(y^2) = 2y\frac{dy}{dx}, \qquad \frac{d}{dx}(\sin y) = \cos y\frac{dy}{dx}, \qquad \frac{d}{dx}(xy) = x\frac{dy}{dx} + y.$$

Then collect $\frac{dy}{dx}$ terms on one side and solve.

Worked Example: 1 (Easy) — $\sqrt{x} + \sqrt{y} = \sqrt{a}$ [Board Feb 2026 Q9, 2 marks]

Differentiating w.r.t. x ($\sqrt{a}$ is a constant):

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}}\frac{dy}{dx} = 0 \quad\Rightarrow\quad \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}} = -\sqrt{\frac{y}{x}}$$

Worked Example: 2 (Easy) — $x\sqrt{x} + y\sqrt{y} = a\sqrt{a}$ [Board Jul 2025 Q9, 2 marks]

Rewrite: $x^{3/2} + y^{3/2} = a^{3/2}$.

$$\frac{3}{2}x^{1/2} + \frac{3}{2}y^{1/2}\frac{dy}{dx} = 0 \quad\Rightarrow\quad \frac{dy}{dx} = -\frac{\sqrt{x}}{\sqrt{y}} = -\sqrt{\frac{x}{y}}$$

Worked Example: 3 (Medium) — $x^3 + x^2y + xy^2 + y^3 = 81$

$$3x^2 + \left(x^2\frac{dy}{dx} + 2xy\right) + \left(x\cdot 2y\frac{dy}{dx} + y^2\right) + 3y^2\frac{dy}{dx} = 0$$

$$\frac{dy}{dx}\left(x^2 + 2xy + 3y^2\right) = -\left(3x^2 + 2xy + y^2\right) \quad\Rightarrow\quad \frac{dy}{dx} = -\frac{3x^2 + 2xy + y^2}{x^2 + 2xy + 3y^2}$$

Worked Example: 4 (Medium) — $e^{x+y} = \cos(x-y)$

$$e^{x+y}\left(1 + \frac{dy}{dx}\right) = -\sin(x-y)\left(1 - \frac{dy}{dx}\right)$$

$$e^{x+y} + e^{x+y}\frac{dy}{dx} = -\sin(x-y) + \sin(x-y)\frac{dy}{dx}$$

$$\frac{dy}{dx}\left[e^{x+y} - \sin(x-y)\right] = -\left[e^{x+y} + \sin(x-y)\right] \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{\cos(x-y)+\sin(x-y)}{\cos(x-y)-\sin(x-y)}$$

(using $e^{x+y} = \cos(x-y)$ to tidy the answer).

Worked Example: 5 (Medium) — $\log(x+y) = xy$, find $\frac{dy}{dx}$ at $x = 0$ [Board Jul 2025 Q1(v) MCQ]

First find y when $x = 0$: $\log(0+y) = 0 \Rightarrow y = 1$.
Differentiate: $\frac{1}{x+y}\left(1+\frac{dy}{dx}\right) = x\frac{dy}{dx} + y$.
At $(0,1)$: $\frac{1}{1}\left(1+\frac{dy}{dx}\right) = 0+1 \Rightarrow \frac{dy}{dx} = 0$. Option (d).
(Textbook Misc. Ex. version: $\log(x+y) = 2xy$ gives $1+y' = 2$, so $y'(0) = 1$.)

Worked Example: 6 (Medium) — $x = e^{x/y}$ [Board Jul 2024 Q1(v) MCQ]

Make it explicit first. Taking log: $\log x = \frac{x}{y} \Rightarrow y = \frac{x}{\log x}$.

$$\frac{dy}{dx} = \frac{\log x \cdot 1 - x \cdot \frac{1}{x}}{(\log x)^2} = \frac{\log x - 1}{(\log x)^2}$$

The options are in terms of x and y. Since $\log x = \frac{x}{y}$:

$$\frac{\frac{x}{y}-1}{\frac{x^2}{y^2}} = \frac{y(x-y)}{x^2} = \frac{x-y}{x\cdot\frac{x}{y}} = \frac{x-y}{x\log x} \qquad \text{Option (c)}$$

8.2 Special results (Ex. 1.3 Q4, Q5) — know them, but show working

Formula: Result 1: $x^m y^n = (x+y)^{m+n} \Rightarrow \frac{dy}{dx} = \frac{y}{x}$

Proof for $x^7y^5 = (x+y)^{12}$: taking log,

$$7\log x + 5\log y = 12\log(x+y)$$

$$\frac{7}{x} + \frac{5}{y}\frac{dy}{dx} = \frac{12}{x+y}\left(1+\frac{dy}{dx}\right)$$

$$\frac{dy}{dx}\left(\frac{5}{y} - \frac{12}{x+y}\right) = \frac{12}{x+y} - \frac{7}{x}$$

$$\frac{dy}{dx}\cdot\frac{5x+5y-12y}{y(x+y)} = \frac{12x-7x-7y}{x(x+y)} \Rightarrow \frac{dy}{dx}\cdot\frac{5x-7y}{y} = \frac{5x-7y}{x} \Rightarrow \frac{dy}{dx} = \frac{y}{x}$$

Textbook also notes: for such curves $\frac{d^2y}{dx^2} = 0$ (because $y = cx$, a straight line through the origin).

Formula: Result 2: $F!\left(\frac{x^n \pm y^n}{x^n \mp y^n}\right) = \text{constant} \Rightarrow \frac{dy}{dx} = \frac{y}{x}$

Example: $\sec\left(\frac{x^5+y^5}{x^5-y^5}\right) = a^2$. Then $\frac{x^5+y^5}{x^5-y^5} = \sec^{-1}(a^2) = k$ (a constant). By componendo–

dividendo,

$$\frac{2x^5}{2y^5} = \frac{k+1}{k-1} \Rightarrow \frac{y^5}{x^5} = \frac{k-1}{k+1} = c^5 \Rightarrow y = cx \Rightarrow \frac{dy}{dx} = c = \frac{y}{x}.$$

Don't differentiate the sec directly — the algebra explodes.

Worked Example: 7 (Hard) — infinite radical: $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \cdots \infty}}}$

The part under the first root after $\tan x$ is again y (removing one term from an infinite chain leaves it unchanged).

$$y = \sqrt{\tan x + y} \Rightarrow y^2 = \tan x + y$$

$$2y\frac{dy}{dx} = \sec^2 x + \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{\sec^2 x}{2y-1}$$

Worked Example: 8 (Hard) — $x^y = e^{x-y}$, show $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$

Taking log: $y \log x = x - y \Rightarrow y(1+\log x) = x \Rightarrow y = \dfrac{x}{1+\log x}$.

$$\frac{dy}{dx} = \frac{(1+\log x)\cdot 1 - x \cdot \frac{1}{x}}{(1+\log x)^2} = \frac{\log x}{(1+\log x)^2}$$

Lesson: if y can be made explicit after taking log, do it — it's shorter.

Worked Example: 9 (Hard) — $y = x^{x^{x^{\cdots^{\infty}}}}$, show $\frac{dy}{dx} = \frac{y^2}{x(1-\log y)}$

$y = x^y \Rightarrow \log y = y \log x$.

$$\frac{1}{y}\frac{dy}{dx} = y \cdot \frac{1}{x} + \log x \frac{dy}{dx} \Rightarrow \frac{dy}{dx}\left(\frac{1}{y} - \log x\right) = \frac{y}{x} \Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1-y\log x)}$$

But $y \log x = \log y$, so $\dfrac{dy}{dx} = \dfrac{y^2}{x(1-\log y)}$.

Common Mistake: Implicit differentiation errors

- Missing $\frac{dy}{dx}$ on a y term, especially inside products like x^2y (needs product rule: $x^2y' + 2xy$).
- Differentiating a constant like a^2, $\sqrt{a}$ or 81 to something non-zero.
- For "value at a point" questions, find the missing coordinate first (Example 5).

9 1.4 Parametric Functions

9.1 Idea

Sometimes x and y are both given in terms of a third variable (a **parameter**), e.g. $x = a\cos t$, $y = a\sin t$ (a circle traced as t changes). Think of t as time: $x(t)$ is the horizontal position and $y(t)$ the vertical position of a moving point. The slope of its path is (vertical speed) ÷ (horizontal speed).

Theorem: 1.4.2 Derivative of parametric functions

If $x = f(t)$ and $y = g(t)$ are differentiable functions of t so that y is a function of x, and $\frac{dx}{dt} \neq 0$, then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}.$$

9.2 Proof — the most-asked proof of this chapter (Mar 2024, Feb 2025, Jul 2025, Jun 2026: all 4-mark)

Proof: Write it exactly in this order

Given: $x = f(t)$ and $y = g(t)$ are differentiable functions of t, and y is a function of x.
Let δt be a small increment in t, and δx, δy the corresponding increments in x and y, where $\delta t \neq 0$, $\delta x \neq 0$, $\delta y \neq 0$.
Consider

$$\frac{\delta y}{\delta x} = \frac{\frac{\delta y}{\delta t}}{\frac{\delta x}{\delta t}}$$

Taking the limit as $\delta t \to 0$ on both sides,

$$\lim_{\delta t \to 0} \frac{\delta y}{\delta x} = \frac{\lim\limits_{\delta t \to 0} \frac{\delta y}{\delta t}}{\lim\limits_{\delta t \to 0} \frac{\delta x}{\delta t}}$$

As $\delta t \to 0$, $\delta x \to 0$ ($\because x = f(t)$ is a differentiable, hence continuous, function of t).

$$\therefore \lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{\lim\limits_{\delta t \to 0} \frac{\delta y}{\delta t}}{\lim\limits_{\delta t \to 0} \frac{\delta x}{\delta t}} \qquad \text{.....(I)}$$

Since x and y are differentiable functions of t,

$$\lim_{\delta t \to 0} \frac{\delta y}{\delta t} = \frac{dy}{dt} \quad \text{and} \quad \lim_{\delta t \to 0} \frac{\delta x}{\delta t} = \frac{dx}{dt} \neq 0 \qquad \text{.....(II)}$$

From (I) and (II),

$$\lim_{\delta x \to 0} \frac{\delta y}{\delta x} = \frac{dy/dt}{dx/dt} \qquad \text{.....(III)}$$

Since $\frac{dx}{dt} \neq 0$, the R.H.S. of (III) exists and is finite. Hence the L.H.S. exists and is finite, i.e. y is a differentiable function of x, and

$$\boxed{\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}, \quad \frac{dx}{dt} \neq 0}$$

Exam Tip: Marking points in the parametric proof

(1) increments with non-zero conditions; (2) the ratio $\frac{\delta y/\delta t}{\delta x/\delta t}$; (3) **"$\delta x \to 0$ as $\delta t \to 0$ since x is continuous"**; (4) identifying limits with $\frac{dx}{dt} \neq 0$ and concluding. The proof is worth ~2–3 of the 4 marks; the "hence" sum is ~1–2.

9.3 Worked examples (including every "hence" part the board has asked)

Worked Example: 1 (Easy) — $x = at^2, y = 2at$ [Board Mar 2024 Q31 hence]

$$\frac{dx}{dt} = 2at, \qquad \frac{dy}{dt} = 2a \qquad \Rightarrow \qquad \frac{dy}{dx} = \frac{2a}{2at} = \frac{1}{t}$$

Worked Example: 2 (Easy) — $y = at^2, x = 2at$ [Board Jul 2025 Q31 hence]

Note the swap. $\frac{dy}{dt} = 2at$, $\frac{dx}{dt} = 2a$, so $\frac{dy}{dx} = \frac{2at}{2a} = t$.

Worked Example: 3 (Easy) — $x = \sin t, y = \cos t$ [Board Jun 2026 Q31 hence]

$$\frac{dx}{dt} = \cos t, \quad \frac{dy}{dt} = -\sin t \quad \Rightarrow \quad \frac{dy}{dx} = \frac{-\sin t}{\cos t} = -\tan t \quad (\cos t \neq 0)$$

Worked Example: 4 (Medium) — slope at a point: $x = \sin\theta, y = \cos 2\theta$ at $\theta = \frac{\pi}{6}$ [Board Mar 2024 Q1(v)]

$$\frac{dy}{dx} = \frac{-2\sin 2\theta}{\cos\theta} = \frac{-4\sin\theta\cos\theta}{\cos\theta} = -4\sin\theta$$

At $\theta = \frac{\pi}{6}$: $-4 \cdot \frac{1}{2} = -2$. Option (c).

Worked Example: 5 (Medium) — $x = a\cos^3\theta, y = a\sin^3\theta$ at $\theta = \frac{\pi}{3}$ [Ex. 1.4 Q2(ii)]

$$\frac{dx}{d\theta} = -3a\cos^2\theta\sin\theta, \qquad \frac{dy}{d\theta} = 3a\sin^2\theta\cos\theta$$

$$\frac{dy}{dx} = \frac{3a\sin^2\theta\cos\theta}{-3a\cos^2\theta\sin\theta} = -\tan\theta \Rightarrow \left(\frac{dy}{dx}\right)_{\theta=\pi/3} = -\sqrt{3}$$

Worked Example: 6 (Hard) — inverse trig parametric: $x = \cos^{-1}\left(\frac{2t}{1+t^2}\right), y = \sec^{-1}\sqrt{1+t^2}$ [Ex. 1.4 Q1(vii)]

Put $t = \tan\theta$ (take $0 < \theta < \frac{\pi}{4}$, i.e. $0 < t < 1$).

$x = \cos^{-1}(\sin 2\theta) = \cos^{-1}\left[\cos\left(\frac{\pi}{2} - 2\theta\right)\right] = \frac{\pi}{2} - 2\theta$, so $\frac{dx}{d\theta} = -2$.

$y = \sec^{-1}(\sec\theta) = \theta$, so $\frac{dy}{d\theta} = 1$.

$$\frac{dy}{dx} = \frac{1}{-2} = -\frac{1}{2}$$

9.4 1.4.3 Derivative of one function with respect to another

To differentiate $u(x)$ **w.r.t.** $v(x)$, treat x as the parameter:

$$\frac{du}{dv} = \frac{du/dx}{dv/dx}, \qquad \frac{dv}{dx} \neq 0.$$

Worked Example: 1 (Easy) — 7^x w.r.t. x^7 [Board Feb 2025 Q31 hence; textbook Solved Ex. 1]

Let $u = 7^x$, $v = x^7$. $\frac{du}{dx} = 7^x \log 7$, $\frac{dv}{dx} = 7x^6$.

$$\frac{du}{dv} = \frac{7^x \log 7}{7x^6} = \frac{7^{x-1}\log 7}{x^6}$$

Worked Example: 2 (Medium) — $\cos^{-1} x$ w.r.t. $\sqrt{1-x^2}$

$u = \cos^{-1} x \Rightarrow \frac{du}{dx} = -\frac{1}{\sqrt{1-x^2}}$; $v = \sqrt{1-x^2} \Rightarrow \frac{dv}{dx} = \frac{-x}{\sqrt{1-x^2}}$.

$$\frac{du}{dv} = \frac{-1/\sqrt{1-x^2}}{-x/\sqrt{1-x^2}} = \frac{1}{x}$$

Worked Example: 3 (Hard) — $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ w.r.t. $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$, $0 < x < 1$

Put $x = \tan\theta$, $0 < \theta < \frac{\pi}{4}$. $u = \sin^{-1}(\sin 2\theta) = 2\theta = 2\tan^{-1} x$ and $v = \cos^{-1}(\cos 2\theta) = 2\theta = 2\tan^{-1} x$. So $u = v$ and $\frac{du}{dv} = 1$.

Common Mistake: Parametric errors

- Writing $\frac{dx/dt}{dy/dt}$ — upside down. y **on top always**, because we want $\frac{dy}{dx}$.
- Forgetting the condition $\frac{dx}{dt} \neq 0$ in proofs.
- In "w.r.t." questions, students differentiate one function w.r.t. the other directly. Always go via x.

10 1.5 Higher Order Derivatives

10.1 Definition and notation

If $y = f(x)$, then $\frac{dy}{dx}$ is again a function of x; differentiating it again gives the **second order derivative**:

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = f''(x) = y_2 = y''.$$

Similarly $\frac{d^3y}{dx^3} = y_3, \ldots, \frac{d^ny}{dx^n} = y_n$.

Physical meaning. If s is displacement at time t: $\frac{ds}{dt}$ = velocity, $\frac{d^2s}{dt^2}$ = acceleration. (Board example: $s = 2t^3 - 5t^2 + 4t - 3$ gives $a = 12t - 10$.)

Polynomial fact (textbook Note). If $f(x)$ is a polynomial of degree n, then $f^{(n)}(x)$ is a constant and all higher derivatives are zero.

Worked Example: 1 (Easy) — $y = x^3 + 7x^2 - 2x - 9$ [Board Feb 2026 Q10, 2 marks]

$$\frac{dy}{dx} = 3x^2 + 14x - 2, \qquad \frac{d^2y}{dx^2} = 6x + 14$$

Worked Example: 2 (Medium) — implicit second derivative: $x^2 + y^2 = a^2$

$2x + 2yy_1 = 0 \Rightarrow y_1 = -\dfrac{x}{y}$.

$$y_2 = -\frac{y \cdot 1 - x\,y_1}{y^2} = -\frac{y - x\left(-\frac{x}{y}\right)}{y^2} = -\frac{y^2 + x^2}{y^3} = -\frac{a^2}{y^3}$$

Technique: after differentiating y_1, **substitute** y_1 **back** and then use the original equation to simplify.

Worked Example: 3 (Medium) — parametric second derivative: $x = at^2, y = 2at$

From earlier, $\dfrac{dy}{dx} = \dfrac{1}{t}$. Now

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{t}\right) = \frac{d}{dt}\left(\frac{1}{t}\right) \cdot \frac{dt}{dx} = -\frac{1}{t^2} \cdot \frac{1}{2at} = -\frac{1}{2at^3}$$

Common Mistake: The #1 error in parametric second derivatives

$\dfrac{d^2y}{dx^2} \neq \dfrac{d^2y/dt^2}{d^2x/dt^2}$, and $\dfrac{d^2y}{dx^2} \neq \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)$. You must multiply by $\dfrac{dt}{dx} = \dfrac{1}{dx/dt}$. In the example above, the wrong method gives $-\frac{1}{t^2}$ instead of $-\frac{1}{2at^3}$.

10.2 "Show that" (differential equation) questions

Strategy. Find y_1; **clear denominators/roots by cross-multiplying** (and squaring if needed); then differentiate again. This avoids quotient-rule mess.

Worked Example: 4 (Medium) — $y = \sin^{-1} x$, show $(1 - x^2)y_2 - xy_1 = 0$ [Board Mar 2024 Q21, 3 marks]

$$y_1 = \frac{1}{\sqrt{1 - x^2}} \;\Rightarrow\; \sqrt{1 - x^2}\,y_1 = 1$$

Differentiating w.r.t. x:

$$\sqrt{1 - x^2}\,y_2 + y_1 \cdot \frac{-2x}{2\sqrt{1 - x^2}} = 0$$

Multiplying by $\sqrt{1 - x^2}$:

$$(1 - x^2)y_2 - xy_1 = 0$$

Worked Example: 5 (Hard) — $y = \cos(m \cos^{-1} x)$, show $(1 - x^2)y_2 - xy_1 + m^2 y = 0$ [textbook Solved Ex. 4]

$$y_1 = -\sin(m\cos^{-1} x) \cdot m \cdot \frac{-1}{\sqrt{1 - x^2}} = \frac{m \sin(m \cos^{-1} x)}{\sqrt{1 - x^2}}$$

$$\sqrt{1 - x^2}\,y_1 = m \sin(m \cos^{-1} x)$$

Squaring: $(1 - x^2)y_1^2 = m^2 \sin^2(m \cos^{-1} x) = m^2[1 - \cos^2(m \cos^{-1} x)] = m^2(1 - y^2)$.
Differentiating w.r.t. x:

$$(1 - x^2) \cdot 2y_1 y_2 + y_1^2(-2x) = m^2(-2yy_1)$$

Dividing by $2y_1$ ($y_1 \neq 0$):

$$(1-x^2)y_2 - xy_1 = -m^2y \;\Rightarrow\; (1-x^2)y_2 - xy_1 + m^2y = 0$$

Worked Example: 6 (Hard) — $y = A\cos(\log x) + B\sin(\log x)$, show $x^2y_2 + xy_1 + y = 0$ [Misc. Ex. 7(iv)]

$$y_1 = -\frac{A\sin(\log x)}{x} + \frac{B\cos(\log x)}{x} \;\Rightarrow\; xy_1 = -A\sin(\log x) + B\cos(\log x)$$

Differentiating again:

$$xy_2 + y_1 = -\frac{A\cos(\log x)}{x} - \frac{B\sin(\log x)}{x} = -\frac{y}{x}$$

Multiplying by x: $x^2y_2 + xy_1 + y = 0$.

10.3 1.5.2 n^{th} derivatives of standard functions

Method. Find y_1, y_2, y_3, spot the pattern, write y_n.

Worked Example: n^{th} derivative of $\log x$ [Board Feb 2025 Q21, 3 marks]

$$y = \log x$$

$$y_1 = \frac{1}{x} = x^{-1}$$

$$y_2 = (-1)x^{-2} = \frac{(-1)^1\,1!}{x^2}$$

$$y_3 = (-1)(-2)x^{-3} = \frac{(-1)^2\,2!}{x^3}$$

$$y_4 = (-1)(-2)(-3)x^{-4} = \frac{(-1)^3\,3!}{x^4}$$

In general,

$$\boxed{y_n = \frac{(-1)^{n-1}(n-1)!}{x^n}}$$

Check: $n = 1$ gives $\frac{(-1)^0 0!}{x} = \frac{1}{x}$ ✓.

Worked Example: n^{th} derivative of $\sin(ax+b)$

$y_1 = a\cos(ax+b) = a\sin\left(ax+b+\frac{\pi}{2}\right)$ (since $\cos\theta = \sin(\theta+\frac{\pi}{2})$).
$y_2 = a^2\cos\left(ax+b+\frac{\pi}{2}\right) = a^2\sin\left(ax+b+\frac{2\pi}{2}\right)$. Each differentiation multiplies by a and adds $\frac{\pi}{2}$:

$$y_n = a^n\sin\left(ax+b+\frac{n\pi}{2}\right)$$

Formula: Standard n^{th} derivatives (Ex. 1.5 Q4)

y	y_n
$(ax+b)^m$	$m(m-1)\cdots(m-n+1)\,a^n(ax+b)^{m-n}$

$\dfrac{1}{ax+b}$	$\dfrac{(-1)^n\, n!\, a^n}{(ax+b)^{n+1}}$
e^{ax+b}	$a^n e^{ax+b}$
a^{px+q}	$p^n (\log a)^n\, a^{px+q}$
$\log(ax+b)$	$\dfrac{(-1)^{n-1}(n-1)!\, a^n}{(ax+b)^n}$
$\sin(ax+b)$	$a^n \sin\left(ax+b+\frac{n\pi}{2}\right)$
$\cos(ax+b)$	$a^n \cos\left(ax+b+\frac{n\pi}{2}\right)$
$e^{ax}\sin(bx+c)$	$r^n e^{ax}\sin(bx+c+n\alpha)$, $r=\sqrt{a^2+b^2}$, $\alpha=\tan^{-1}\frac{b}{a}$
$e^{ax}\cos(bx+c)$	$r^n e^{ax}\cos(bx+c+n\alpha)$

Worked Example: Using the table — $y=\cos(3-2x)$

Here $a=-2$, $b=3$: $\quad y_n=(-2)^n\cos\left(3-2x+\frac{n\pi}{2}\right)$.

Worked Example: $e^{ax}\cos(bx+c)$ — derive y_1 to see where r and α come from

$y_1 = ae^{ax}\cos(bx+c) - be^{ax}\sin(bx+c) = e^{ax}[a\cos(bx+c) - b\sin(bx+c)]$. Put $a = r\cos\alpha$, $b = r\sin\alpha$ (so $r=\sqrt{a^2+b^2}$, $\tan\alpha=\frac{b}{a}$):

$$y_1 = re^{ax}[\cos(bx+c)\cos\alpha - \sin(bx+c)\sin\alpha] = re^{ax}\cos(bx+c+\alpha)$$

Repeating, $y_n = r^n e^{ax}\cos(bx+c+n\alpha)$. E.g. $y=e^{8x}\cos(6x+7)$: $r=10$, $\alpha=\tan^{-1}\frac{3}{4}$, $y_n = 10^n e^{8x}\cos\left(6x+7+n\tan^{-1}\frac{3}{4}\right)$.

Common Mistake: n^{th} derivative errors

- Sign pattern: for $\log x$ the sign is $(-1)^{n-1}$, for $\frac{1}{x}$ it is $(-1)^n$. Check with $n=1$.
- Factorial: $\log x$ has $(n-1)!$; $\frac{1}{x}$ has $n!$.
- Forgetting the a^n from the inner function $ax+b$.

11 Exam Toolkit

If you see…	Use

11.1 Which method? (decision guide)

If you see…	Use
Function inside a function	Chain rule (1.1)
"derivative of inverse", $(f^{-1})'(k)$	$\frac{dx}{dy} = \frac{1}{dy/dx}$; find x with $f(x) = k$ first
$\sin^{-1}$, $\tan^{-1}$ of a simple inside	Direct formula $\times$ inner derivative
$\tan^{-1}/\sin^{-1}/\cos^{-1}$ of $\frac{2x}{1\pm x^2}$, $\sqrt{\frac{a-x}{a+x}}$, $3x-4x^3$, $\frac{\cos x}{1+\sin x}$, $\frac{a\cos x+b\sin x}{\sqrt{a^2+b^2}}$	Trig substitution, simplify, then differentiate
$\tan^{-1}\frac{(p+q)x}{1-pqx^2}$	Split into $\tan^{-1} px + \tan^{-1} qx$
Variable in base **and** power	Logarithmic differentiation
Long product / quotient	Logarithmic differentiation
Sum of two f^g terms	Split $y = u + v$, log each separately
Equation mixing x and y	Implicit differentiation
$x^m y^n = (x+y)^{m+n}$ or $F\left(\frac{x^n \pm y^n}{x^n \mp y^n}\right) = c$	Answer is $\frac{y}{x}$; show log / componendo steps
Infinite nested radical / power	Replace the tail by y
$x = f(t)$, $y = g(t)$	$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$
"Differentiate u w.r.t. v"	$\frac{du/dx}{dv/dx}$
"Show that $(\dots)y_2 + (\dots)y_1 + (\dots)y = 0$"	Find y_1, cross-multiply/square, differentiate again
"n^{th} derivative"	Find $y_1, y_2, y_3 \to$ pattern, or use table

11.2 Presentation tips for full marks

- Begin with "Given: …" or "Let $y = \dots$". Write "Differentiating w.r.t. x, we get" before each differentiation.
- In proofs, number the equations (I), (II), (III) as the textbook does and state the continuity reason.
- When you simplify with a substitution, write the substitution and the range of θ.
- Box or underline the final answer. In MCQs, write the option letter **and** the answer (some papers award zero for letter only).

12 Final Summary

Summary: Complete formula list for Chapter 1

Chain rule: $\dfrac{dy}{dx} = \dfrac{dy}{du}\cdot\dfrac{du}{dx}$; $\dfrac{d}{dx}f[g(x)] = f'[g(x)]\,g'(x)$.

Inverse function: $\dfrac{dx}{dy} = \dfrac{1}{dy/dx}$; $(f^{-1})'(y) = \dfrac{1}{f'(x)}$; $g'(x) = \dfrac{1}{f'[g(x)]}$.

Inverse trig: $(\sin^{-1}x)' = \dfrac{1}{\sqrt{1-x^2}}$, $(\cos^{-1}x)' = \dfrac{-1}{\sqrt{1-x^2}}$, $(\tan^{-1}x)' = \dfrac{1}{1+x^2}$, $(\cot^{-1}x)' = \dfrac{-1}{1+x^2}$, $(\sec^{-1}x)' = \dfrac{1}{x\sqrt{x^2-1}}$, $(\operatorname{cosec}^{-1}x)' = \dfrac{-1}{x\sqrt{x^2-1}}$.

Log differentiation: $\dfrac{d}{dx}f^g = f^g\left[\dfrac{g}{f}f' + g'\log f\right]$.

Parametric: $\dfrac{dy}{dx} = \dfrac{dy/dt}{dx/dt}$; $\dfrac{d^2y}{dx^2} = \dfrac{d}{dt}\left(\dfrac{dy}{dx}\right)\cdot\dfrac{dt}{dx}$.

One w.r.t. another: $\dfrac{du}{dv} = \dfrac{du/dx}{dv/dx}$.

Special implicit: $x^m y^n = (x+y)^{m+n} \Rightarrow y' = \dfrac{y}{x}$, $y'' = 0$.

Inverse second derivative (Misc. 5(vi)): $\dfrac{d^2x}{dy^2} = -\left(\dfrac{dy}{dx}\right)^{-3}\dfrac{d^2y}{dx^2}$.

n^{th} derivatives: see table in 1.5.2; especially $(\log x)_n = \dfrac{(-1)^{n-1}(n-1)!}{x^n}$, $\left(\frac{1}{x}\right)_n = \dfrac{(-1)^n n!}{x^{n+1}}$.

13 Self-Assessment (try first, answers on the next page)

1. Differentiate $y = \sqrt{\cos\sqrt{x}}$ w.r.t. x.
2. If $y = \tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$, $|x|<1$, find $\dfrac{dy}{dx}$.
3. If $f(x) = x^3 + x - 2$, find $(f^{-1})'(0)$.
4. Differentiate $y = x^{\sin x}$.
5. If $e^x + e^y = e^{x+y}$, show that $\dfrac{dy}{dx} = -e^{y-x}$.
6. If $x = a(1-\cos\theta)$, $y = b(\theta - \sin\theta)$, find $\dfrac{dy}{dx}$.
7. Differentiate $\tan^{-1}x$ w.r.t. $\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$, $x>0$.
8. If $y = e^{m\tan^{-1}x}$, show that $(1+x^2)y_2 + (2x-m)y_1 = 0$.
9. Find the nth derivative of $\dfrac{1}{3x-5}$.
10. If $y = \log\left[\dfrac{x+\sqrt{x^2+a^2}}{\sqrt{x^2+a^2}-x}\right]$, find $\dfrac{dy}{dx}$. (Board Jun 2026 Q21)
11. **Proof:** State and prove the chain rule; hence find $\dfrac{d}{dx}\left(\dfrac{1}{\sqrt{\sin x}}\right)$. (Board Jul 2024 Q31)
12. If $\sin y = x\sin(a+y)$, show that $\dfrac{dy}{dx} = \dfrac{\sin^2(a+y)}{\sin a}$.

13.1 Answers with key steps

1. $\dfrac{dy}{dx} = \dfrac{1}{2\sqrt{\cos\sqrt{x}}} \cdot (-\sin\sqrt{x}) \cdot \dfrac{1}{2\sqrt{x}} = -\dfrac{\sin\sqrt{x}}{4\sqrt{x}\sqrt{\cos\sqrt{x}}}$.

2. $x = \tan\theta$, $\theta \in (-\frac{\pi}{4}, \frac{\pi}{4})$: $y = \tan^{-1}(\tan 2\theta) = 2\tan^{-1} x$, so $\dfrac{dy}{dx} = \dfrac{2}{1+x^2}$.

3. $f(x) = 0 \Rightarrow x = 1$ ($1 + 1 - 2 = 0$). $f'(x) = 3x^2 + 1$, $f'(1) = 4$. $(f^{-1})'(0) = \dfrac{1}{4}$.

4. $\log y = \sin x \log x$; $\dfrac{dy}{dx} = x^{\sin x}\left[\cos x \log x + \dfrac{\sin x}{x}\right]$.

5. $e^x + e^y y' = e^{x+y}(1 + y') \Rightarrow y'(e^y - e^{x+y}) = e^{x+y} - e^x$. Using $e^{x+y} = e^x + e^y$: $e^y - e^{x+y} = -e^x$ and $e^{x+y} - e^x = e^y$. So $y' = \dfrac{e^y}{-e^x} = -e^{y-x}$.

6. $\dfrac{dx}{d\theta} = a\sin\theta$, $\dfrac{dy}{d\theta} = b(1 - \cos\theta)$. $\dfrac{dy}{dx} = \dfrac{b(1-\cos\theta)}{a\sin\theta} = \dfrac{b \cdot 2\sin^2\frac{\theta}{2}}{a \cdot 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}} = \dfrac{b}{a}\tan\dfrac{\theta}{2}$.

7. $v = 2\tan^{-1} x$ (for $x > 0$), $u = \tan^{-1} x$. $\dfrac{du}{dv} = \dfrac{1/(1+x^2)}{2/(1+x^2)} = \dfrac{1}{2}$.

8. $y_1 = \dfrac{my}{1+x^2} \Rightarrow (1+x^2)y_1 = my$. Differentiate: $(1+x^2)y_2 + 2xy_1 = my_1 \Rightarrow (1+x^2)y_2 + (2x - m)y_1 = 0$.

9. $a = 3$, $b = -5$: $y_n = \dfrac{(-1)^n n!\, 3^n}{(3x-5)^{n+1}}$.

10. Multiply numerator and denominator by $\sqrt{x^2+a^2} + x$: $\dfrac{(x + \sqrt{x^2+a^2})^2}{(x^2+a^2) - x^2} = \dfrac{(x+\sqrt{x^2+a^2})^2}{a^2}$. So $y = 2\log(x + \sqrt{x^2+a^2}) - 2\log a$. $\dfrac{dy}{dx} = \dfrac{2}{x+\sqrt{x^2+a^2}}\left(1 + \dfrac{x}{\sqrt{x^2+a^2}}\right) = \dfrac{2}{x+\sqrt{x^2+a^2}} \cdot \dfrac{\sqrt{x^2+a^2}+x}{\sqrt{x^2+a^2}} = \dfrac{2}{\sqrt{x^2+a^2}}$.

11. Proof as in Section 1.1. Hence: $y = (\sin x)^{-1/2}$, $\dfrac{dy}{dx} = -\dfrac{1}{2}(\sin x)^{-3/2}\cos x = -\dfrac{\cos x}{2\sin x\sqrt{\sin x}} = -\dfrac{1}{2}\cot x\sqrt{\operatorname{cosec} x}$.

12. $x = \dfrac{\sin y}{\sin(a+y)}$. Differentiate w.r.t. y: $\dfrac{dx}{dy} = \dfrac{\sin(a+y)\cos y - \sin y\cos(a+y)}{\sin^2(a+y)} = \dfrac{\sin(a+y-y)}{\sin^2(a+y)} = \dfrac{\sin a}{\sin^2(a+y)}$. Hence $\dfrac{dy}{dx} = \dfrac{\sin^2(a+y)}{\sin a}$.

Note: Coverage and limitations

- Textbook source: the Part II textbook PDF in your project (Chapter 1, pp. 1–64). Its text layer drops most fractions, so a few exercise expressions were reconstructed from the surrounding text and standard results; every worked answer here was re-derived and checked independently.
- PYQ source: the six board papers in your project (Mar 2024 – Jun 2026). No marking schemes were in the folder, so mark splits mentioned are estimates based on the question structure, not official schemes.